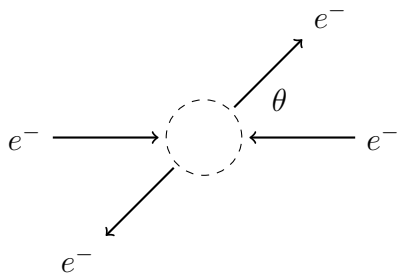
Moller scattering 1

Moller scattering is the process $e^- + e^- \to e^- + e^-$.



Momentum vectors where $E = \sqrt{p^2 + m^2}$.

$$p_1 = \underset{e^- \longrightarrow}{\begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}} \qquad p_2 = \underset{\longleftarrow e^-}{\begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}} \qquad p_3 = \underset{\nearrow}{\overset{e^-}{\begin{pmatrix} E \\ p\sin\theta\cos\phi \\ p\sin\theta\sin\phi \\ p\cos\theta \end{pmatrix}}} \qquad p_4 = \underset{e^- \swarrow}{\begin{pmatrix} E \\ -p\sin\theta\cos\phi \\ -p\sin\theta\sin\phi \\ -p\cos\theta \end{pmatrix}}$$

Spinors for p_1.

$$u_{11} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ p \\ 0 \end{pmatrix}} \qquad u_{12} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ 0 \\ -p \end{pmatrix}}$$

Spinors for p_2.

$$u_{21} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ -p \\ 0 \end{pmatrix}} \qquad u_{22} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ 0 \\ p \end{pmatrix}}$$

Spinors for p_3.

$$u_{31} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ p_{3z} \\ p_{3x} + ip_{3y} \end{pmatrix}} \qquad u_{32} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ p_{3x} - ip_{3y} \\ -p_{3z} \end{pmatrix}}$$

Spinors for p_4.

$$u_{41} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ p_{4z} \\ p_{4x} + ip_{4y} \end{pmatrix}} \qquad u_{42} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ p_{4x} - ip_{4y} \\ -p_{4z} \end{pmatrix}}$$

The scattering amplitude $\mathcal{M}$ is

$$\mathcal{M} = \mathcal{M}_1 - \mathcal{M}_2$$

where $\mathcal{M}_1$ is the $1 \to 3$, $2 \to 4$ amplitude

$$\mathcal{M}_1 = \frac{e^2}{t}(\bar{u}_{4d}\gamma^\mu u_{2b})(\bar{u}_{3c}\gamma_\mu u_{1a})$$

and $\mathcal{M}_2$ is the $1 \to 4$, $2 \to 3$ amplitude

$$\mathcal{M}_2 = \frac{e^2}{u}(\bar{u}_{3c}\gamma^\nu u_{2b})(\bar{u}_{4d}\gamma_\nu u_{1a})$$

The expected density $\langle|\mathcal{M}|^2\rangle$ is the average over spin states.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{abcd}|\mathcal{M}|^2 = \frac{1}{4}\sum_{abcd}(\mathcal{M}_1 - \mathcal{M}_2)(\mathcal{M}_1 - \mathcal{M}_2)^*$$

Expand the summand.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{abcd}(\mathcal{M}_1\mathcal{M}_1^* - \mathcal{M}_1\mathcal{M}_2^* - \mathcal{M}_2\mathcal{M}_1^* + \mathcal{M}_2\mathcal{M}_2^*)$$

The summation results are

$$\sum_{abcd}\mathcal{M}_1\mathcal{M}_1^* = \frac{e^4}{t^2}\left(8s^2 + 8u^2 - 64m^2s - 64m^2u + 192m^4\right)$$
$$\sum_{abcd}\mathcal{M}_1\mathcal{M}_2^* = \frac{e^4}{tu}\left(-8s^2 + 64m^2s - 96m^4\right)$$
$$\sum_{abcd}\mathcal{M}_2\mathcal{M}_2^* = \frac{e^4}{u^2}\left(8s^2 + 8t^2 - 64m^2s - 64m^2t + 192m^4\right)$$

where s, t, and u are Mandelstam variables

$$s = (p_1 + p_2)^2$$
$$t = (p_1 - p_3)^2$$
$$u = (p_1 - p_4)^2$$

For clarity define

$$f_{11} = 8s^2 + 8u^2 - 64m^2s - 64m^2u + 192m^4$$
$$f_{12} = -8s^2 + 64m^2s - 96m^4$$
$$f_{22} = 8s^2 + 8t^2 - 64m^2s - 64m^2t + 192m^4$$

and write $\langle|\mathcal{M}|^2\rangle$ more compactly as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(\frac{f_{11}}{t^2} - \frac{2f_{12}}{tu} + \frac{f_{22}}{u^2}\right)$$

For $E \gg m$ a useful approximation is to set $m = 0$ and obtain

$$f_{11} = 8s^2 + 8u^2$$
$$f_{12} = -8s^2$$
$$f_{22} = 8s^2 + 8t^2$$

Then $\langle|\mathcal{M}|^2\rangle$ simplifies as

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{s^2 + u^2}{t^2} + \frac{2s^2}{tu} + \frac{s^2 + t^2}{u^2} \right)$$

For $m = 0$ the Mandelstam variables are

$$s = 4E^2$$
$$t = -2E^2(1 - \cos\theta)$$
$$u = -2E^2(1 + \cos\theta)$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{1 + \cos^4(\theta/2)}{\sin^4(\theta/2)} + \frac{2}{\sin^2(\theta/2)\cos^2(\theta/2)} + \frac{1 + \sin^4(\theta/2)}{\cos^4(\theta/2)} \right)$$

This can be written more compactly as

$$\langle|\mathcal{M}|^2\rangle = 4e^4 \frac{(\cos^2\theta + 3)^2}{\sin^4\theta}$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{e^4} \frac{\langle|\mathcal{M}|^2\rangle}{4s} = \frac{\alpha^2}{4E^2} \frac{(\cos^2\theta + 3)^2}{\sin^4\theta}$$

Multiply by $(\hbar c)^2$ to convert from natural units to physical units.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E^2} \frac{(\cos^2\theta + 3)^2}{\sin^4\theta} (\hbar c)^2$$

Zee p. 134 has the cross section formula

$$\frac{d\sigma}{d\Omega} = \left(\frac{e^2}{4\pi} \right)^2 \frac{1}{8E^2} f(\theta)$$

where $f(\theta)$ is the function

$$f(\theta) = \frac{1 + \cos^4(\theta/2)}{\sin^4(\theta/2)} + \frac{2}{\sin^2(\theta/2)\cos^2(\theta/2)} + \frac{1 + \sin^4(\theta/2)}{\cos^4(\theta/2)}$$

Function $f(\theta)$ is equivalent to

$$f(\theta) = 2\,\frac{(\cos^2\theta + 3)^2}{\sin^4\theta}$$

Noting that in natural units $e^2 = 4\pi\alpha$ we have

$$\left(\frac{e^2}{4\pi} \right)^2 = \alpha^2$$

Hence

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E^2} \frac{(\cos^2\theta + 3)^2}{\sin^4\theta}$$

Eigenmath script